

Why a night planner

Turning a night of observing from recalled expertise into a reproducible plan

CHIME/FRB host spectroscopy · Keck I / LRIS · Shane 3m / Kast

J. Xavier Prochaska & Claude

Why we need it

Four problems, only one of which is about saving time



Planning a night eats a day

Twilight, LST, airmass windows, hour-angle limits, slews, exposure times — all recomputed by hand for every run, every date, every change of plan.



The constraints live in your head

Keck I's Nasmyth deck vignettes a fixed azimuth band. Shane's hour angle wraps asymmetrically. Masks need weeks of lead time. Forget one and the plan looks fine but cannot be executed.



Observing is easy; planning is not

A well-made plan lets someone with no experience of the instrument run a productive night. The expertise should sit in the plan, not in the person holding it.



Queue and ToO work needs plans on demand

Non-classical modes need many candidate plans, replanned as conditions change. That is only practical if a plan costs seconds rather than an afternoon.

What it refuses to forget

Every one of these silently produced a wrong-but-plausible plan at least once

Keck I Nasmyth deck	az 5.3–146.2° needs alt ≥ 33.3°; rising targets in the NE/E must climb before they clear. Keck II is blocked on the opposite side.
Hour-angle / wrap limits	Shane: −5.0h to +3.75h, asymmetric, and a mount limit rather than a horizon one. Keck: ±5.5h.
Declination limits	Shane cannot go above +82°, which is precisely what makes those targets Keck opportunities.
Mask lead time	A mag > 21 host needs a pre-milled slitmask — weeks ahead, not on the night.
Real exposure times	A magnitude-bucket heuristic carried from another telescope was wrong by ~15x. Exposures now come from the live LRIS ETC.
Slew and overheads	10 min base, 20 min crossing Dec +20 — enough to decide whether a target fits its window.

What it does

From any table of targets to a plan you can observe from

Any target table

FFFF-PZ pulls or your own spreadsheet. Columns are detected automatically and it refuses to guess when a guess could point the telescope elsewhere.

Real exposure times

Live UCO/Lick ETC per target: S/N 10 per exposure, 3 exposures minimum. MOS blocks budgeted by magnitude.

Schedules around reality

Manual start times for a late handback, blocked windows for a ToO, and forced partial exposures when a short spectrum beats none.

Finders and starlists

Image cutout, Gaia offset star, and the slit drawn at the dispersion-weighted parallactic angle — no external service required.

Shows its reasoning

Every target carries the stage it was accepted or rejected at, with numbers. Airmass tracks, idle time, twilight marks.

Browser front end

Pick instrument and date, load targets, generate, inspect, export the whole run as a zip.

Nights it has planned

Including one replanned mid-night around a ToO

Lick-2026B-01

Shane / Kast · Aug 13–14

6 targets

7.0h science, no deadtime

Lick-2026B-02 N1

Shane / Kast · Sep 14

4 of 12

250 / 517 min (48%)

Lick-2026B-02 N2

Shane / Kast · Sep 15–16

6 repeats + 2 spares

402 min science (77%)

Keck-2026B-02

Keck I / LRIS · Oct 6

9 targets

6.9h on target (77%)



A ToO took the first 50 minutes of Oct 6

Replanned from a 20:00 handback in seconds. One target lost to its closing window; a mag 25.6 host that had been dropped was forced in for a partial 80-minute exposure as it rose into the best airmass of the night. On target went 336 → 416 min, 62% → 77%, idle down to 8 min.

Beyond classical observing

The same machinery that replans a ToO plans a queue



Plans as a commodity

When a plan costs seconds, you can generate one per candidate date and compare, instead of defending the first one you built.



Replan on interruption

Manual start times and blocked windows mean a ToO, a weather hold or an instrument fault is a parameter, not a restart.



Feasibility before submission

Answer "is this programme observable in this semester, and for how long" with real windows rather than an estimate.



Hand the night over

A plan carrying its own reasoning, finders and starlist can be executed by someone who has never used the instrument.

Where it goes next

The hard parts are already instrument-independent



DEIMOS on Keck II

The multi-object case in earnest: mask design drives the plan rather than following it. Keck II also vignettes on the opposite side to Keck I — the geometry is already parameterised, so it is a configuration, not a rewrite.



More instruments at the same telescopes

KCWI, ESI, HIRES. Each needs its own exposure model and overheads; everything else — twilight, windows, scheduling, finders, starlists — is shared.



More telescopes

The site, pointing limits and timezone are already data rather than code. A new observatory is a site entry plus an exposure calculator.



Generalise the Lick path

Keck takes any target table and any date today; the Lick pipeline is still tied to one run. Bringing it level is the next real piece of work.



Queue-mode tooling

Rank and re-rank a target pool across many nights, not just schedule one.